

LAKSHMI NARAYANAN GIRIDHARAN - AI/ML Engineer

Chicago, IL | lakshmigiridharan1261@gmail.com | (773) 842-4388 | [LinkedIn](#) | [GitHub](#)

SUMMARY

AI/ML Engineer with 4+ years of global experience in India and U.S., delivering machine learning, natural language processing, and large language model solutions. Skilled in Python, SQL, TensorFlow, PyTorch, Hugging Face, and MLOps tools (AWS, GCP, Azure). Proven success improving classification accuracy 10–17%, reducing pipeline runtime 25–35%, and deploying large-scale applications with stable low-latency APIs. Adept at data engineering, recommendation systems, and LLM fine-tuning for real-world automation. Strong collaborator in Agile teams, recognized for measurable business impact and ability to transform complex datasets into actionable insights driving strategic decision-making.

SKILLS

Methodologies & IDEs: Agile, SDLC, Jupyter, VS Code, PyCharm, Google Colab

Programming & Databases: Python, SQL, C++, R, MATLAB, MySQL, PostgreSQL, MongoDB, DynamoDB, Snowflake.

Libraries & Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, NumPy, Pandas, OpenCV, NLTK, SpaCy, Hugging Face Transformers (BERT, GPT, DistilBERT, Whisper ASR), Matplotlib, Seaborn

Cloud & MLOps Tools: AWS (SageMaker, EC2, S3, Lambda, CloudWatch), Azure (Functions, Blob Storage, ML Studio), GCP (BigQuery, Vertex AI, Cloud Storage), Docker, Kubernetes, MLflow, Airflow, Jenkins, GitHub Actions, VMware

Big Data & Analytics: Apache Spark, Hadoop, Hive, Redshift, Tableau, Power BI, Grafana

Machine Learning & AI: Regression, Classification, Clustering, Random Forest, Decision Trees, Time-Series Forecasting, Deep Learning (CNN, RNN, LSTM), NLP, Recommendation Systems, GANs, LLMs, Prompt Engineering, A/B Testing1

Other Skills: Data Cleaning, Data Wrangling, Data Scrapping, Feature Engineering, Data Visualization, Model Deployment, Experiment Tracking, REST APIs, Version Control (Git)

EXPERIENCE

UnitedHealth Group – Chicago, IL

Jan 25 - Present

AI/ML Engineer

- Deployed fine-tuned LLMs on AWS SageMaker Endpoints, reducing API latency 24% while handling 12K+ monthly queries without reported downtime.
- Developed prompt-engineered GPT pipelines for automated report generation, reducing documentation preparation time 58% and improving stakeholder turnaround cycles for compliance deliverables.
- Designed recommendation engine with Spark and Redshift, processing 120K+ daily interactions and raising product engagement 22% through personalized experiences.
- Automated ingestion pipelines with Airflow and BigQuery, decreasing feature preparation latency 36% and enabling near real-time model retraining in production.
- Delivered Tableau dashboards integrating LLM-driven insights, improving executive decision making efficiency 19% and reducing dependency on ad hoc reporting teams.

Schneider Electric – Tamilnadu, India

Feb 22 - Jul 23

Machine Learning Engineer

- Fine-tuned Hugging Face Transformer models, improving contextual accuracy 15% and lowering hallucinations 18% for automated compliance document summarization pipelines.
- Automated nightly Airflow workflows with Spark, reducing ETL job duration 32% and enabling daily model retraining cycles for predictive analytics solutions.
- Designed recommendation system with Snowflake and Python, delivering personalized suggestions for 90K+ daily users and boosting click-through rates 20% across product categories.
- Built interactive Power BI dashboards visualizing classification results, reducing reporting turnaround time 25% and improving decision-making efficiency for cross-functional business stakeholders.
- Improved computer vision workflows using OpenCV and CNNs, achieving 14% better defect detection accuracy and reducing manual inspection requirements 40% across manufacturing datasets.
- Engineered and deployed end-to-end ML pipelines with scikit-learn and TensorFlow, optimizing hyperparameters to boost model accuracy by 21% and reduce training time 30%.

Python Developer

- Implemented Python pipelines with Scikit-learn, optimizing pre-processing and hyperparameters, improving model prediction accuracy 12% and reducing training runtime 23% across multiple production datasets.
- Automated data cleaning for 600K+ structured records using Pandas and SQL, decreasing pre-processing errors 34%, enhancing downstream model stability, and enabling reliable iterative ML model retraining.
- Built supervised TensorFlow models detecting fraudulent transactions, increasing recall 11%, reducing false negatives, and improving risk assessment confidence for financial reporting teams while minimizing manual review efforts.
- Engineered ETL workflows with Apache Spark and containerized Python models with Docker on AWS EC2, accelerating dataset refresh cycles 28%, decreasing inference latency 21%, and ensuring scalable, reliable ML pipelines.
- Developed Matplotlib / Seaborn visualizations, enabling 18% faster metric interpretation and reducing reliance on manual spreadsheet reports.

EDUCATION

Master of Science - Electrical and Computer Engineering

Aug 23 – May 25

Illinois Institute of Technology, Chicago, IL

Bachelor of Technology - Electrical and Communication Engineering

Mar 18 – Jul 22

SRM Institute of Technology, Chennai, India

PROJECTS

Intelligent Interview Automation Platform [Github.com](#)**Tech Stack:** Java 17 · Spring Boot · Python · FastAPI · Sentence-BERT · DistilBERT · OpenAI Whisper API · Google Cloud TTS · Docker · REST APIs · HTML/CSS/JavaScript.

- Built a voice-interactive interview system using NLP, speech-to-text (Whisper), and semantic/sentiment models (Sentence-BERT, DistilBERT) for automated candidate evaluation.
- Deployed modular microservices via Docker with a real-time web interface for displaying scores, feedback, and downloadable reports.

Impact:

- Reduced manual evaluation time by ~70%.
- Improved assessment accuracy with emotion detection and AI services.

AI-Powered Mortgage Eligibility Assistant [Github.com](#)**Tech Stack:** Python · scikit-learn · FastAPI · OpenAI GPT-4 · ReportLab · Matplotlib · React · AWS EC2 · Docker · REST API.

- Built Scikit-learn classification/regression models to predict loan approval likelihood and refinance savings, and integrated a GPT-4 chatbot via FastAPI for real-time mortgage guidance.
- Developed an interactive React frontend with multi-step forms, visual reports, and deployed the full system on AWS for scalable cloud-based access.

Impact:

- Improved loan prediction accuracy by 23%.
- Boosted query response speed by 42%.
- Increased applicant confidence by 40% through automated PDF eligibility reports.

Real-Time Audio Compliance Monitor [Github.com](#)**Tech Stack:** Whisper ASR · Sentence Transformers · NLP · Streamlit · AWS S3 · Python

- Built an AI system using Whisper for transcription and Sentence-BERT for semantic analysis to detect and redact sensitive speech, integrating secure cloud storage via AWS S3.
- Designed a real-time Streamlit interface supporting live mic input, multi-format uploads, and downloadable redacted reports.

Impact:

- Achieved >90% accuracy in simulated HR compliance scenarios.
- Ensuring high contextual recall and enhancing workplace safety through explainable alerting.

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Microsoft Azure Fundamentals (AZ-900)
- AWS Solutions Architect Associate